

LocalLife — Phase 1 Camera Evaluation

Waste-volume measurement with Intel RealSense D435 (stereo depth) and Logitech C920 (Depth Anything V2 monocular depth)

Data collected 22–29 September 2026 · Status: preliminary, single operator, one room

1. Summary

7.5 % median dimension error (RealSense, 15 objects)	84 % of dimensions within 20 %	57 % / 40 % valid-reading rate RealSense / Logitech	8 / 15 materials correct
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Three data sources are combined here. Only source A has tape-measured ground truth, so it is the only one that supports accuracy statements. Sources B and C compare the two cameras with each other and measure reliability; neither makes an accuracy claim.

Source	Content	Camera(s)	Ground truth	Used for
A. Phase 1 testing sheet	15 household objects, tape-measured L×W×H, material, colour	RealSense D435 only	Yes	Dimension, volume, material and colour accuracy
B. Phase 1 camera comparison (screenshots)	10 scenes, both cameras live on the same objects	Both	No	Qualitative agreement between cameras
C. Automatic log (comparison_measurements.csv)	243 automatic measurement attempts	Both	No	Reliability: accepted, rejected and missing readings

2. Ground-truth accuracy — RealSense D435 (source A)

Method: each object was measured with a tape and then placed under the RealSense. To make the comparison independent of orientation, dimensions are compared sorted (longest with longest). Error colours: green $\leq$ 10 %, yellow $\leq$ 20 %, red $>$ 20 %.

Object	Actual (cm)	Measured (cm)	Err longest %	Err middle %	Err shortest %	Box envelope (L)	Reported (L)	Material (actual → measured)	Colour
Shoes box	36x25x12.5	36x24x12.5	0.0	4.0	0.0	11.25	9.8	paper → cardboard (100%) ✓	black → black
Black polythene bag	30x30x20	28x20x26	6.7	13.3	0.0	18.0	12	plastic → polythene bag (100%) ✓	black → black
Apple waste bag	25x20x8	24x16x10	4.0	20.0	25.0	4.0	2.6	plastic → polythene bag (100%) ✓	transparent → grey
Backpack	35x12x40	42x42x13	5.0	20.0	8.3	16.8	14	fabric → cardboard (75%) ✗	black → black

Polythene bag	26x24x12	25x22x10	3.8	8.3	16.7	7.49	1.4	plastic → polythene bag (100%) ✓	white → grey
Toy	12x10x20	15x9x20	0.0	25.0	10.0	2.4	1.7	fabric → metal (100%) ✗	grey → orange
Brown paper bag	13x30x30	10x28x29	3.3	6.7	23.1	11.7	28	paper → mixed waste (100%) ✗	brown → brown
Drink can	17x7x7	17x6x7	0.0	0.0	14.3	0.83	0.5	metal → fabric (50%) ✗	pink → red
Milk box	9.5x7x22	8.2x6x23.5	6.8	13.7	14.3	1.46	1.1	paper → plastic (100%) ✗	white → blue
Tea box	11x11x17	11x8x17	0.0	0.0	27.3	2.06	0.8	plastic → plastic (100%) ✓	orange → orange
Small plastic box	3.5x3.5x5	4x3x6	20.0	14.3	14.3	0.06	0.07	plastic → plastic (50%) ✓	bluish green → cyan
Plastic cream bottle	8x4x20	8x3x20	0.0	0.0	25.0	0.64	0.3	plastic → plastic (100%) ✓	white → red
Gym equipment	15x12x1.5	12x9x1.4	20.0	25.0	6.7	0.27	0.08	plastic → metal (50%) ✗	green → green
Food waste bag	20x19x4	23x20x5	15.0	5.3	25.0	1.52	0.8	plastic → food waste (100%) ✗	yellow → orange
Pillow	40x40x12	43x38x12	7.5	5.0	0.0	19.2	18	fabric → fabric (100%) ✓	grey → brown

Metric	Result
Median dimension error	7.5 %
Dimensions within 10 % / 20 %	58 % / 84 % (of 45)
Median volume difference vs box envelope	35 % (5/15 within 25 %)
Material correct	8 / 15
Colour exact / close	6 / 15 exact, +5 close

The reported litres come from a height map, which gives the occupied volume. A box envelope overstates the volume of soft or irregular items such as bags, pillows and backpacks. The volume "difference" therefore mixes shape effects with measurement error and should not be read as pure error. Dimensions are the reliable metric here.

3. Paired live readings — RealSense vs Logitech (source B)

The same object was measured at the same moment by both cameras. There is no tape measurement for these objects, so the table shows how far the cameras agree, not which one is correct.

Object	RealSense L×W×H (mm)	RealSense (L)	Logitech L×W×H (mm)	Logitech (L)	Logitech / RealSense volume
Orange can	154×68×170	1.259	95×80×186	0.971	0.77
Filled bag	244×176×100	3.480	295×241×92	2.995	0.86
Backpack	297×188×318	12.925	271×216×302	8.305	0.64
Transparent bag	279×119×162	2.648	356×183×158	4.537	1.71
Milk carton	89×40×216	0.797	97×78×216	1.167	1.46
Tall can	83×47×164	0.470	95×80×97	0.490	1.04

In 4 of the 6 objects the heights agree within about 10 %, and the volumes agree within a factor of about 1.5. Logitech footprints tend to be wider, which is consistent with monocular depth blurring object edges. The tall can (164 vs 97 mm height) shows that monocular depth can fail on narrow, tall objects.

4. Reliability of automatic operation (source C)

	RealSense	Logitech
Measurement attempts	119	124
Accepted	68 (57 %)	50 (40 %)
Rejected with a stated reason	18	28
Missing (no final reading)	33	46
Implausible-size outliers (flagged)	2	1
Usable after flagging	56	40

Main rejection reasons were:

- RealSense: no valid measurement (8), unstable volume (5), candidate spans the measurement zone (3).
- Logitech: no measurable height above the plane (15), no valid measurement (8), unstable volume (5).

The flagged outliers were:

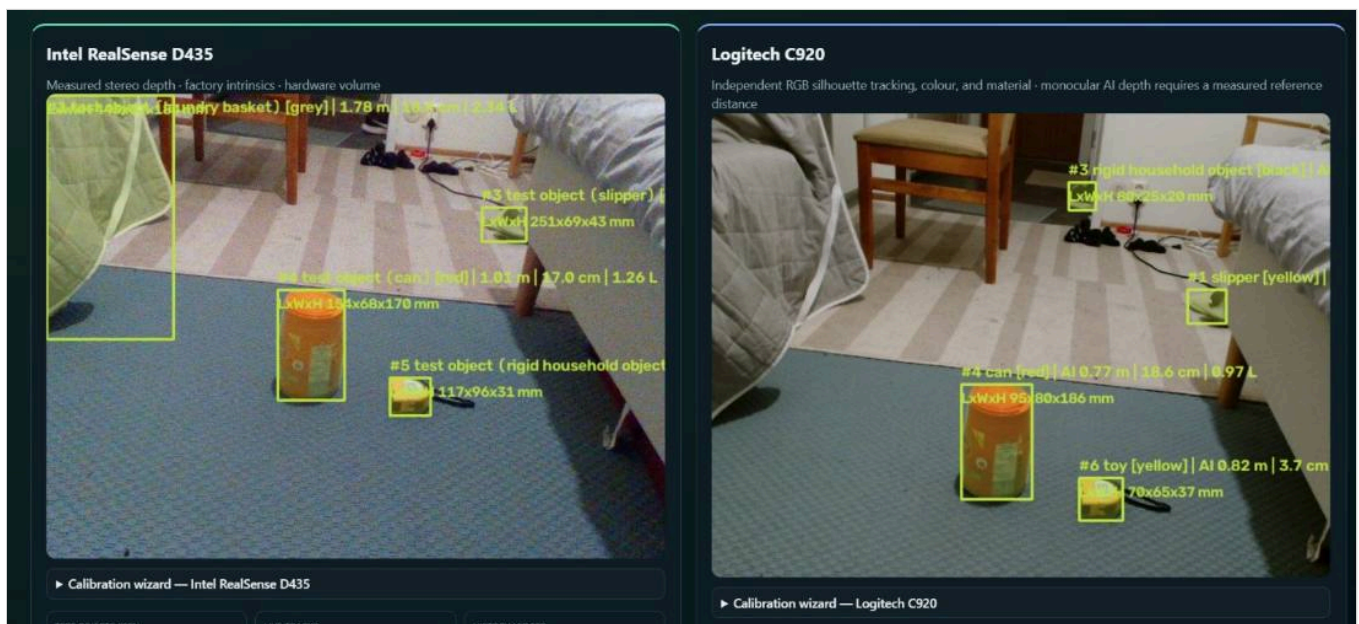
- RealSense: a 3003×538×462 mm (77 L) item and a 1502 mm textile item.
- Logitech: a 1406×695 mm carton.

Both cameras accepted a reading in 15 paired events, but only 5 of these were the same object type in both cameras. Because the time window paired different objects, this log is used for reliability only, not for cross-camera accuracy.

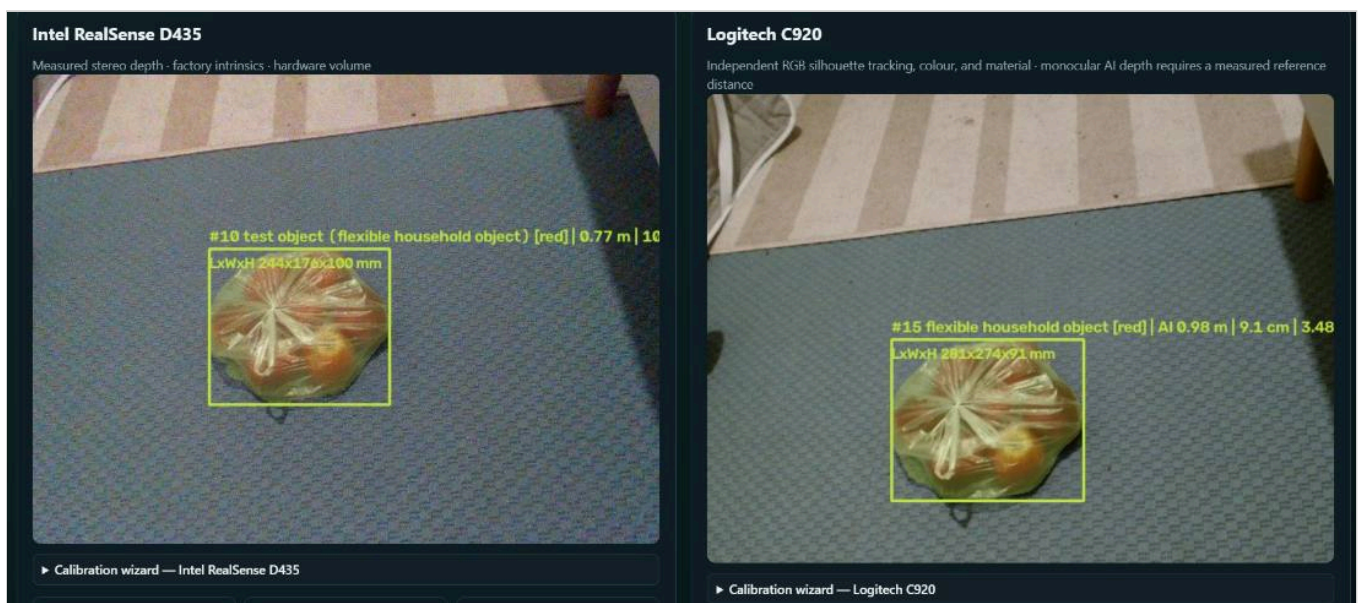
5. Interpretation and limitations

- RealSense dimensions are usable: the median error is 7.5 % and 84 % of dimensions are within 20 %. Volume depends on how "volume" is defined (occupied height map versus box envelope), and the thesis should state this definition explicitly.
- Material classification (8/15 correct) and colour (6/15 exact) are weak. Labels sometimes disagree with obvious appearance (a drink can classified as fabric, a toy as metal).
- The Logitech camera accepts fewer readings (40 %) and depends on a measured reference distance. Its dominant failure is "no measurable height", which means monocular depth did not separate the object from the floor.
- Limitations: 15 ground-truth objects, one room, one operator, a table-level viewpoint. There is no ground truth for the Logitech camera, and the log has no repeated trials per object.
- Next step: repeat source A with both cameras on the same tape-measured objects, 3 trials each, from the final bird's-eye mounting.

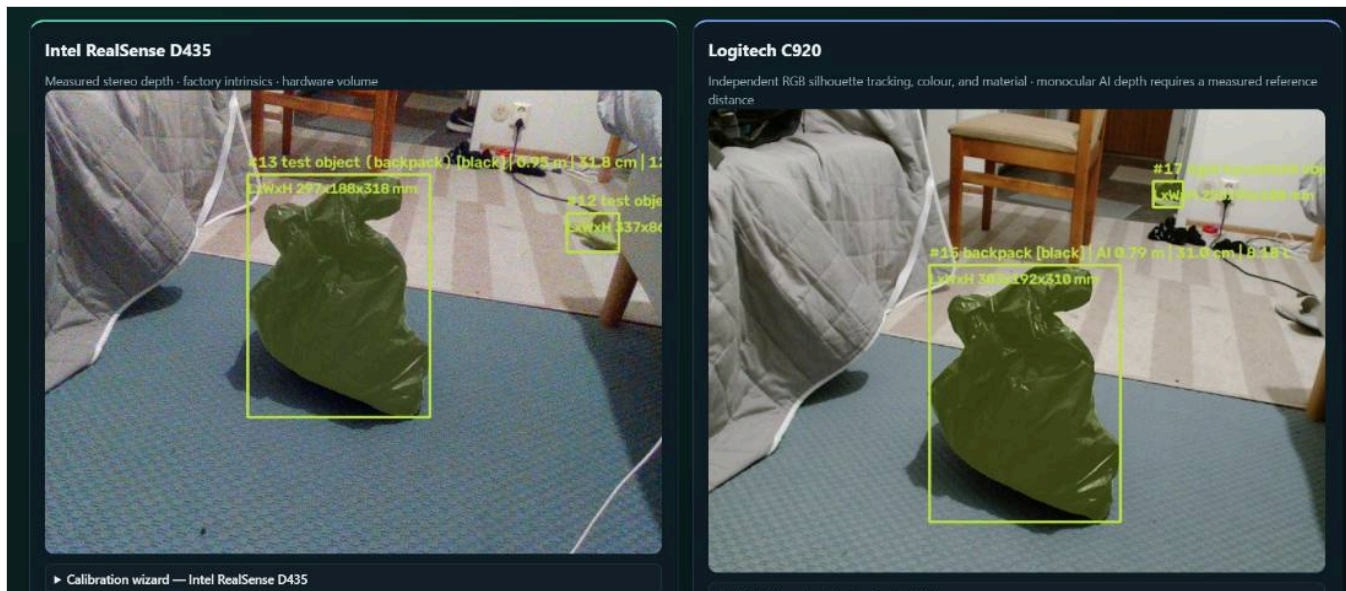
Appendix — Camera comparison screenshots



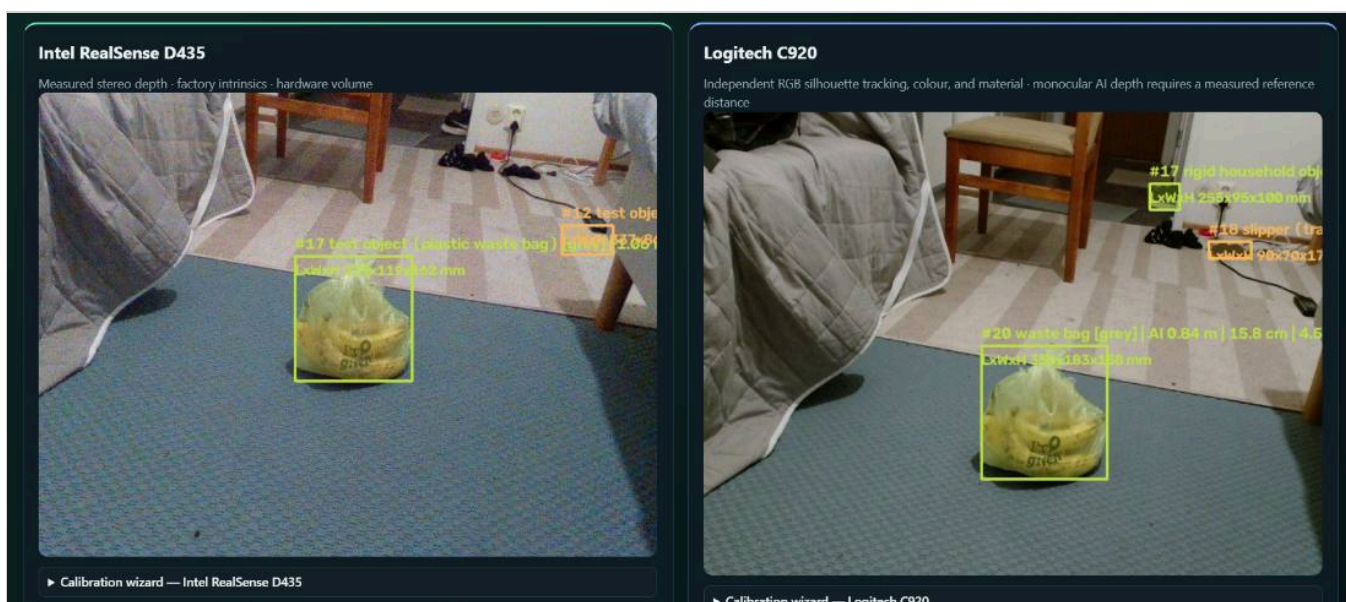
Scene 1 — RealSense (left) vs Logitech (right), live overlay



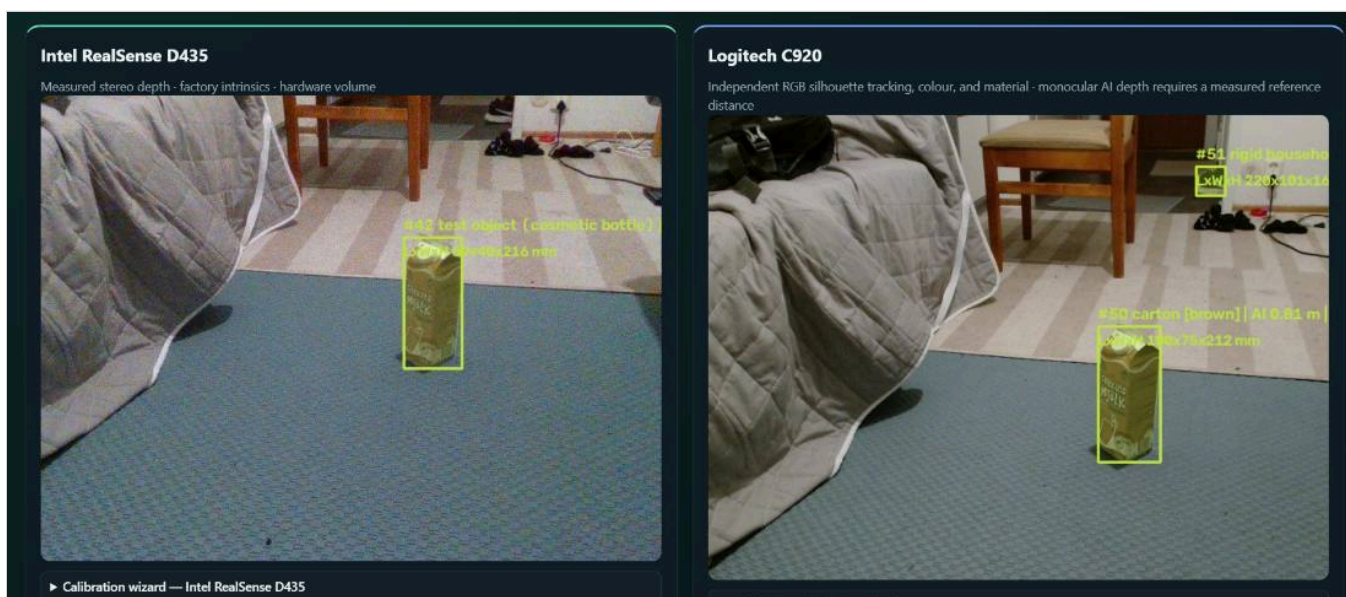
Scene 2 — RealSense (left) vs Logitech (right), live overlay



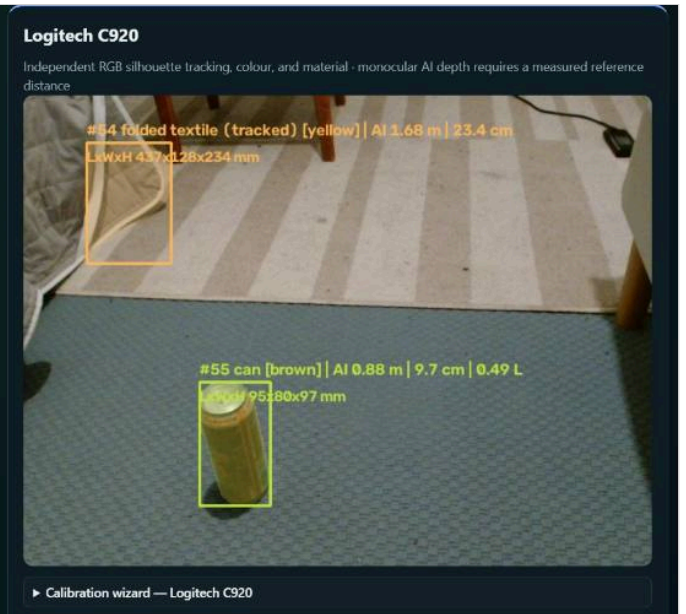
Scene 3 — RealSense (left) vs Logitech (right), live overlay



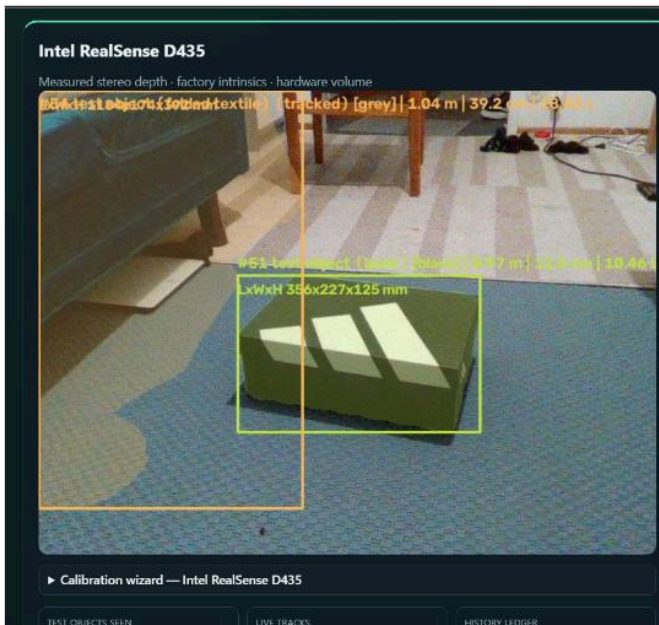
Scene 4 — RealSense (left) vs Logitech (right), live overlay



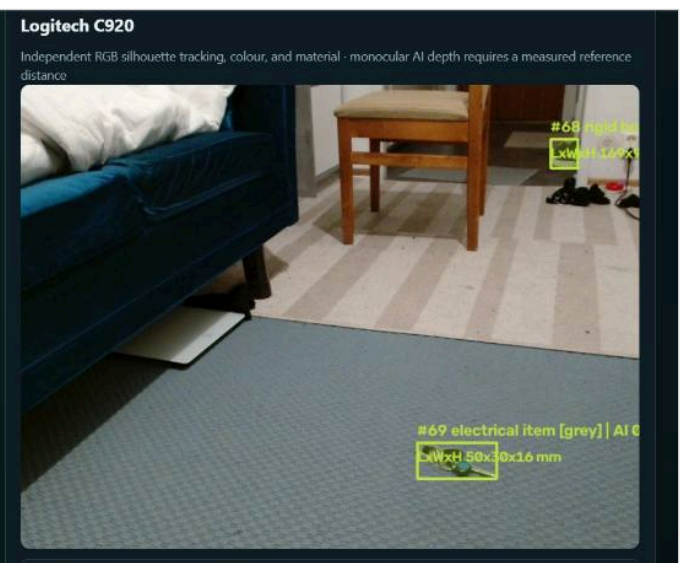
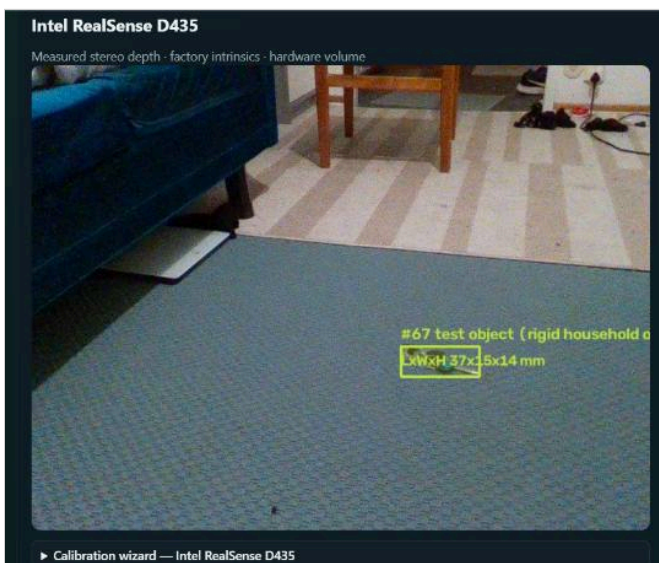
Scene 5 — RealSense (left) vs Logitech (right), live overlay



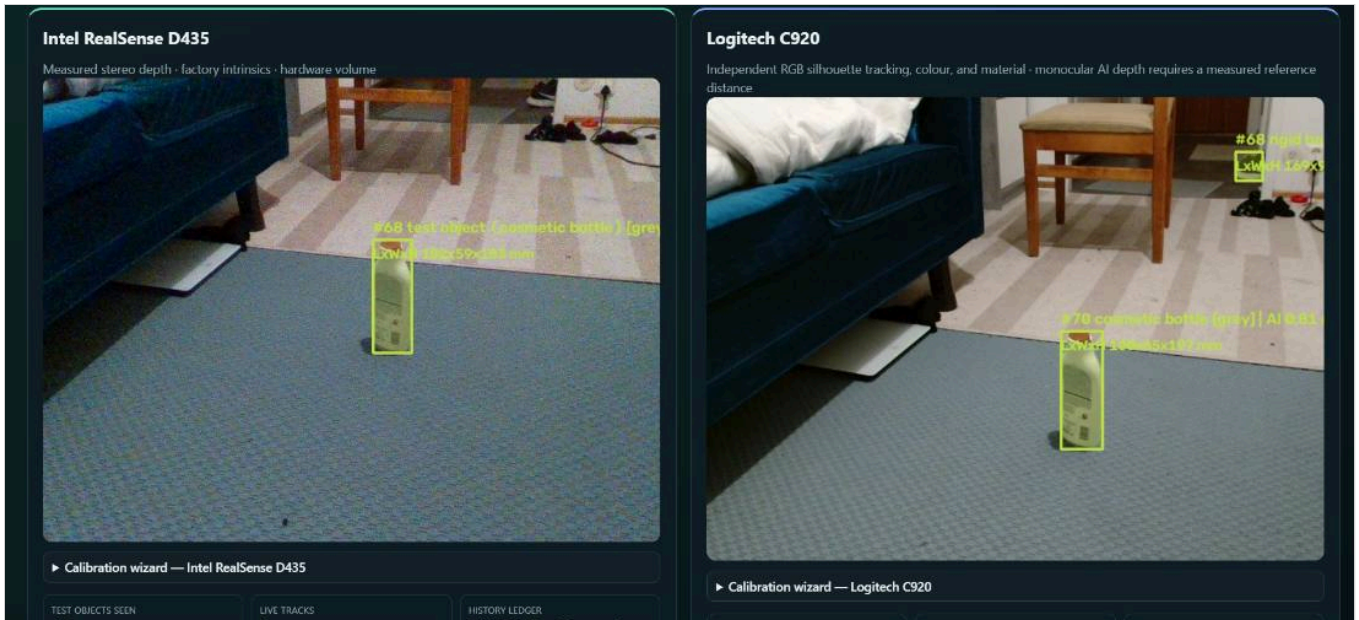
Scene 6 — RealSense (left) vs Logitech (right), live overlay



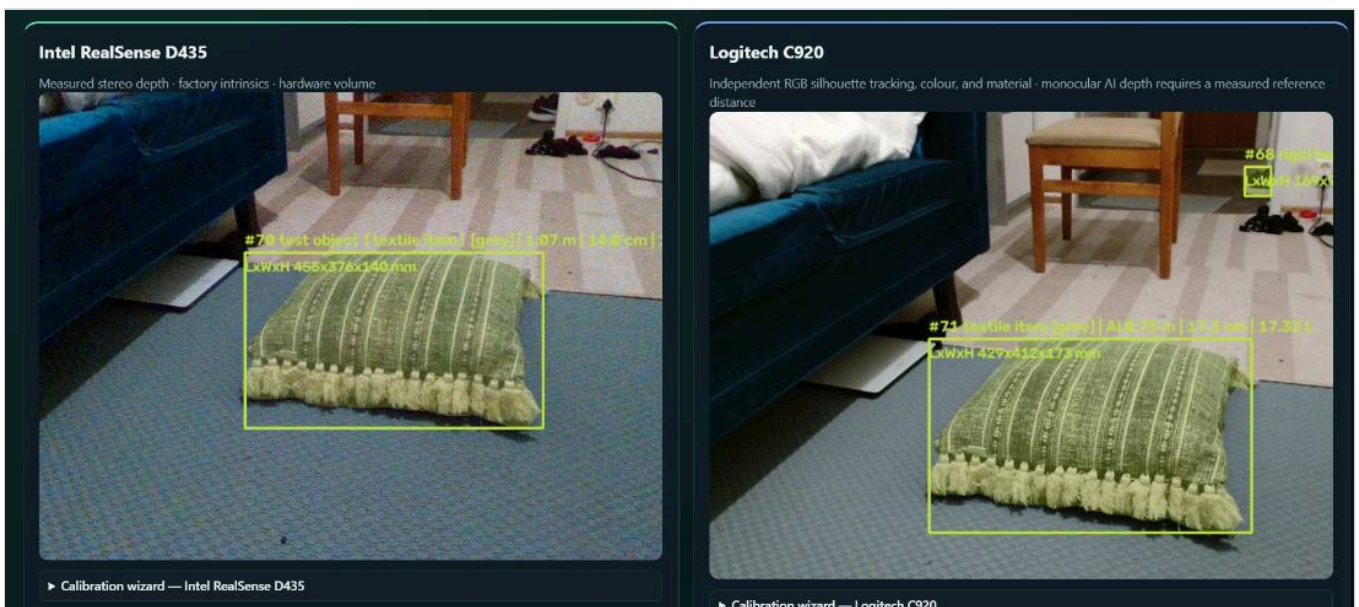
Scene 7 — RealSense (left) vs Logitech (right), live overlay



Scene 8 — RealSense (left) vs Logitech (right), live overlay



Scene 9 — RealSense (left) vs Logitech (right), live overlay



Scene 10 — RealSense (left) vs Logitech (right), live overlay

ID	Accepted type	Color	Sorting	Material	Height	Liters	Added this deposit	RealSense 3D footprint LxWxH
10	test object - flexible household object	red (49%)	UNKNOWN / MANUAL CHECK	polythene bag (72%)	10.0 cm	3.480 L	—	244x176x100 mm 85% conf realSense support plane footprint single_view_visible_footprint, filled_footprint, live_fitted_support_plane depth 100% FINAL d18c8a4b88ee8b87

RealSense history					
Time	Type	Color	Material	Liters	Status
9/28/2026, 9:54:04 PM	rigid household object	mixed	plastic	0.277 L	accepted - History saved: Yes - CSV saved: Yes

ID	Accepted type	Color	Sorting	Material	Height	Liters	Added this deposit	RealSense 3D footprint LxWxH
15	flexible household object	red (44%)	UNKNOWN / MANUAL CHECK	polythene bag (66%)	9.2 cm	2.995 L	—	295x241x92 mm 58% conf logitech floor plane footprint geometry_not_settled_yet, support_surface_not_floor_uncertain depth 58%

Logitech history					
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Example of the per-camera detection tables recorded with each scene